

Resultater fra QH2 for Sunniva Tomren

Resultat for dette forsøket: 6 av 9

Innlevert 3. sep. i 17.38

Dette forsøket tok 90 minutter

Riktig svar



Spørsmål 1

1 / 1 poeng

Hydraulic cylinder, ideal, basic flow/speed relations . The cylinder is retracting . Cylinder diameter (d) = 100 mm . Cylinder rod diameter (d_R) = 50 mm . Cylinder in-flow (Q_i) = 39 l/min . What is the cylinder velocity (v) in mm/s?

110,3474 (med marginen: 1,1035)

Riktig svar



Spørsmål 2

1 / 1 poeng

Hydraulic cylinder, ideal, basic pressure/force relations . The cylinder is retracting . The cylinder is subjected to an assistive (negative) load . Cylinder diameter (d) = 50 mm . Cylinder rod diameter (d_R) = 35 mm . Cylinder load force (F_L) = 7.1 kN . Cylinder piston side pressure (p_1) = 45 bar . What is the cylinder rod side pressure (p_2) in bar?

17,3333 (med marginen: 0,1734)

Riktig svar



Spørsmål 3

1 / 1 poeng

Hydraulic cylinder, basic pressure/force relations . The cylinder is extracting . The cylinder is subjected to a resistive (positive) load . Cylinder diameter (d) = 63 mm . Cylinder rod diameter (d_R) = 40 mm . Cylinder piston side pressure (p_1) = 159 bar . Cylinder rod side pressure (p_2) = 13 bar . Cylinder load force (F_L) = 41.6 kN . What is the cylinder hydromechanical efficiency (η_{hmC})?

0,8824 (med marginen: 0,0088)

Feil svar



Spørsmål 4

0 / 1 poeng

Hydraulic cylinder, buckling, basic relations . Foot mounted, steel rod . Cylinder rod diameter (d_R) = 90 mm . Cylinder base length (L_1) = 0 mm . Cylinder total length (L) = 1243 mm . Cylinder load force (F_L) = 736.2 kN . What is the safety against buckling (S)?

5,8684 (med marginen: 0,0587)

Feil svar



Spørsmål 5

0 / 1 poeng

Hydraulic cylinder, buckling, design considerations . End mounted, steel rod . Cylinder base length (L_1) = 540 mm . Cylinder total length (L) = 686 mm . Safety against buckling (S) = 6 . Cylinder load force (F_L) = 155.9 kN . What is the minimum cylinder rod diameter (d_R) in mm?

30,9775 (med marginen: 0,3098)

Riktig svar



Spørsmål 6

1 / 1 poeng

Accumulator, basic relations . Isothermal process, 1 -> 2 . Initial accumulator volume (V_1) = 16.4 l . Final accumulator volume (V_2) = 25.3 l . Initial accumulator pressure (p_1) = 78 bar . What is the final accumulator pressure (p_2) in bar?

50,5613 (med marginen: 0,5056)

Riktig svar



Spørsmål 7

2 / 2 poeng

Accumulator, design considerations I . Compute accumulator preload (total) volume . Accumulator polytropic coefficient (n_L) = 1.4 . Change in volume (ΔV) = 3.8 l . Accumulator minimum load pressure ($p_{\min}$) = 71 bar . Accumulator maximum load pressure ($p_{\max}$) = 99 bar . Accumulator load pressure (p_L) = 97 bar . Accumulator preload pressure (p_0) = 64 bar . What is the preload (total) accumulator gas volume (V_0) in l?

21,8041 (med marginen: 0,218)

Feil svar



Spørsmål 8

0 / 1 poeng

Accumulator, design considerations II . Accumulator connected to ideal motor . Motor decompressing accumulator in adiabatic process . Motor displacement (D) = $1000 \text{ cm}^3/\text{rev}$. Motor revolutions during process (ΔM) = 13 rev . Final accumulator pressure (p_2) = 63 bar . Initial accumulator pressure (p_1) = 85 bar . Accumulator polytropic coefficient (n_L) = 1.4 . What is the final accumulator gas volume (V_2) in l?

67,4962 (med marginen: 0,6749)

Quizresultat: 6 av 9